

# Predicting Subscription Fatigue and Platform Switching Among OTT Streaming Users

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Sector: Entertainment / OTT streaming services

## 1. Problem Statement and Objectives

Streaming services such as Netflix, Amazon Prime Video and Disney+ Hotstar compete for the same viewers, and leaving costs a subscriber almost nothing: a plan can be paused, downgraded or cancelled in a few taps. Many households therefore hold several subscriptions at once and rotate them as the value they see changes, a pattern known as multi-homing. Jeong and Lee (2025) find that multi-homers who feel subscription fatigue, and who face high fees and high search costs, are more likely to discontinue a service.

In this report, subscription fatigue means the build-up of dissatisfaction with a subscription that comes before a user switches or cancels. It rarely has a single cause. Someone who finds the plan too expensive for how little they watch needs a different response from someone who is happy with the price but tired of buffering, weak recommendations or a thin catalogue. A blanket discount wastes margin on the second person and leaves the real problem untouched.

The business problem is therefore twofold: which subscribers are likely to switch, and what kind of fatigue lies behind it, so that retention spend can be targeted. As agreed in my approved proposal, the prediction target is whether a respondent expects to switch, downgrade or cancel their primary OTT subscription within the next six months.

### Objectives

1. Identify which pricing, content, experience and viewing-behaviour factors are most closely associated with subscription fatigue and switching intention.
2. Diagnose whether each respondent's fatigue is mainly cost-, content- or experience-driven, and measure how common each type is.
3. Build and evaluate an interpretable classification model for six-month switching intention, and turn the results into segment-specific retention recommendations.

## 2. Data Collection and Dataset Description

I collected the primary data with a structured Google Forms questionnaire aimed at OTT subscribers. It has six blocks (Table 1): demographics, subscription details, viewing behaviour, satisfaction and experience ratings, switching history, and a few open-text questions. Satisfaction items use a five-point Likert scale, and the target question asks directly whether the respondent expects to switch, downgrade or cancel their main subscription in the next six months.

Table 1. Questionnaire structure

| Block             | Variables captured                                                        | Purpose                |
|-------------------|---------------------------------------------------------------------------|------------------------|
| Demographics      | Age, gender, occupation, income                                           | Respondent profile     |
| Subscription      | Platform, number of subscriptions, tenure, monthly spend, account sharing | Subscription intensity |
| Viewing behaviour | Weekly viewing hours, preferred genre, main device, usual viewing time    | Usage pattern          |

| Block                     | Variables captured                                                                                                                 | Purpose               |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Satisfaction / experience | Pricing satisfaction, content satisfaction, content variety, recommendation quality, advertisement tolerance, streaming experience | Fatigue indicators    |
| Switching                 | Prior switching, competitor awareness, six-month switching intention                                                               | Target and predictors |
| Open text                 | Reasons for switching, what would encourage staying, suggestions                                                                   | Qualitative context   |

The form produced **13 usable responses**, which is too few to train and test a classifier. To be transparent about what the analysis rests on: the analytical file holds **10,000 records, of which 13 are real survey responses and 9,987 are synthetic**, generated to follow the answer patterns and relationships seen in the real responses. Every row carries a *Data\_Source* label (Real or Synthetic) so the two groups can be separated at any time. Because 99.87% of the rows are synthetic, all percentages, charts and model scores in this report describe the analytical dataset. They do not describe OTT users in general. I treat them as a demonstration of the workflow and as hypotheses to be tested on more genuine responses.

**Table 2. Analytical dataset summary**

| Item                  | Value                                                                                                                                          |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Total records         | 10,000 (32 columns)                                                                                                                            |
| Real survey responses | 13 (0.13%)                                                                                                                                     |
| Synthetic records     | 9,987 (99.87%)                                                                                                                                 |
| Target variable       | Switch_Intent (Yes / No); 9,999 records carry a label                                                                                          |
| Missing values        | Switched_Before 1, Competitor_Awareness 1, Switch_Intent 1, Main_Reason_For_Switching 1, What_Would_Encourage_Stay 1, Suggestions_To_Improve 7 |
| Duplicates            | 0 duplicate rows, 0 duplicate Record_IDs                                                                                                       |

The main variables are the target (*Switch\_Intent*), six 1–5 ratings (pricing satisfaction, content satisfaction, content variety, recommendation quality, advertisement tolerance, streaming experience), subscription and spending details, prior switching, competitor awareness, and four derived fatigue scores (cost, content, experience and overall). No names, e-mail addresses or phone numbers were kept; each respondent appears only as a *Record\_ID*.

### 3. Data Preparation and Exploratory Analysis

#### Preparation steps

- Cleaned column names and category labels so each variable is coded the same way throughout.
- Converted range answers (age, income, tenure, monthly spend, weekly viewing hours) to representative numeric values, stored in the *\_Numeric* columns.
- Kept the six Likert items as 1–5 integers. The Yes/No target was mapped to 0/1; the other Yes/No fields are one-hot encoded inside the modelling pipeline (Section 4).
- Checked for duplicates (none). One record had no target label and was left out of modelling, leaving 9,999. The few other missing values are imputed inside the pipeline.
- Kept open-text answers as background only; they are not model inputs.
- Derived cost, content and experience fatigue scores (higher means more fatigued) from the related ratings, plus an overall score, and assigned each respondent a dominant fatigue type from them. The

correlations in Appendix A1 show how the scores are built: cost fatigue tracks pricing satisfaction ( $-0.90$ ) and monthly spend ( $+0.44$ ), content fatigue tracks content satisfaction ( $-0.73$ ) and variety ( $-0.67$ ), and experience fatigue tracks recommendations, advertisement tolerance and streaming experience ( $-0.57$ ,  $-0.51$ ,  $-0.64$ ).

### Descriptive analysis

The respondent profile is young and student-heavy: 69.4% are students, the median age is 21, and 61.3% are male. Viewing is concentrated at night (61.7%). Respondents hold 2.8 subscriptions on average and spend about ₹545 a month. This narrow profile limits how far the results can be generalised. Netflix is the primary platform for 69.25% of respondents (Figure 1a).

**Table 3. Key distributions**

| Indicator                        | Category                                       | Share of records         |
|----------------------------------|------------------------------------------------|--------------------------|
| Primary platform                 | Netflix / Amazon Prime Video / Disney+ Hotstar | 69.25% / 23.40% / 7.35%  |
| Occupation                       | Student                                        | 69.42%                   |
| Switching intention (six months) | Yes / No                                       | 29.88% / 70.12%          |
| Switched a subscription before   | Yes                                            | 49.88%                   |
| Aware of competitor offers       | Yes                                            | 48.92%                   |
| Dominant fatigue type            | Cost / Experience / Content                    | 44.25% / 34.51% / 21.24% |

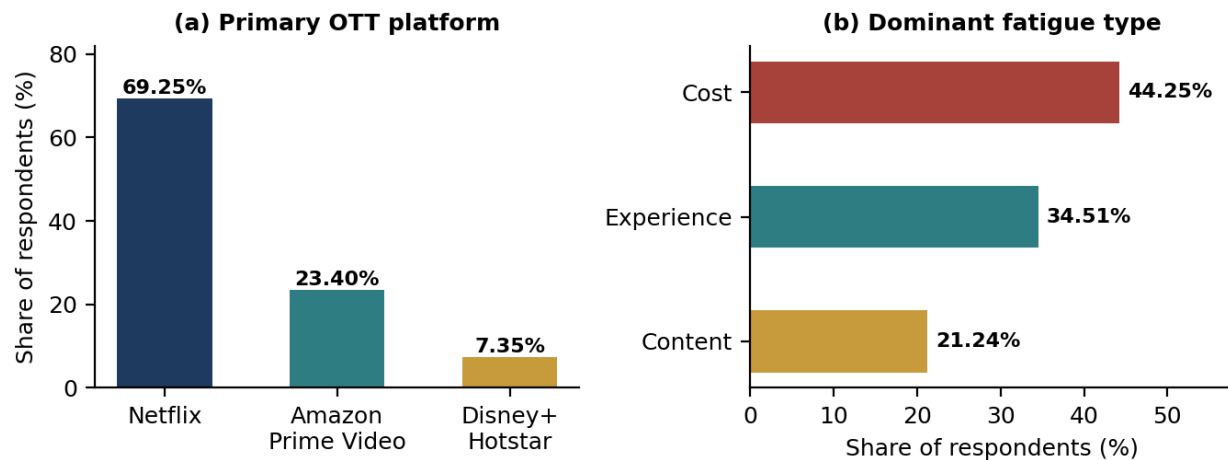


Figure 1. (a) Primary OTT platform and (b) dominant fatigue type, as a share of all 10,000 records.

Cost is the most common dominant fatigue type (44.25%), followed by experience (34.51%) and content (21.24%). Price-to-usage value is therefore the most frequent single source of dissatisfaction, but more than half of respondents (55.75%) are mainly bothered by something else, which a price-only retention plan would miss. These shares cover all respondents, not only those who intend to switch. Since only 29.88% intend to switch, the target is imbalanced, so I judge the model on precision, recall and F1 rather than accuracy alone.

Switching intention falls as satisfaction rises (Figure 2a). The drop is steepest for pricing: from about 36% among respondents who rated pricing 1 to about 25% among those who rated it 5. Content satisfaction follows a similar pattern (about 39% at rating 2 to 28% at rating 5), and streaming experience is flatter (about 33% to 28%). The fatigue scores tell the same story more weakly (Figure 2b): people who intend to switch score about 2.5 points higher on cost fatigue (47.9 against 45.4), 1.3 higher on content fatigue and 0.8 higher on experience fatigue. The gaps are consistent in direction but modest in size.

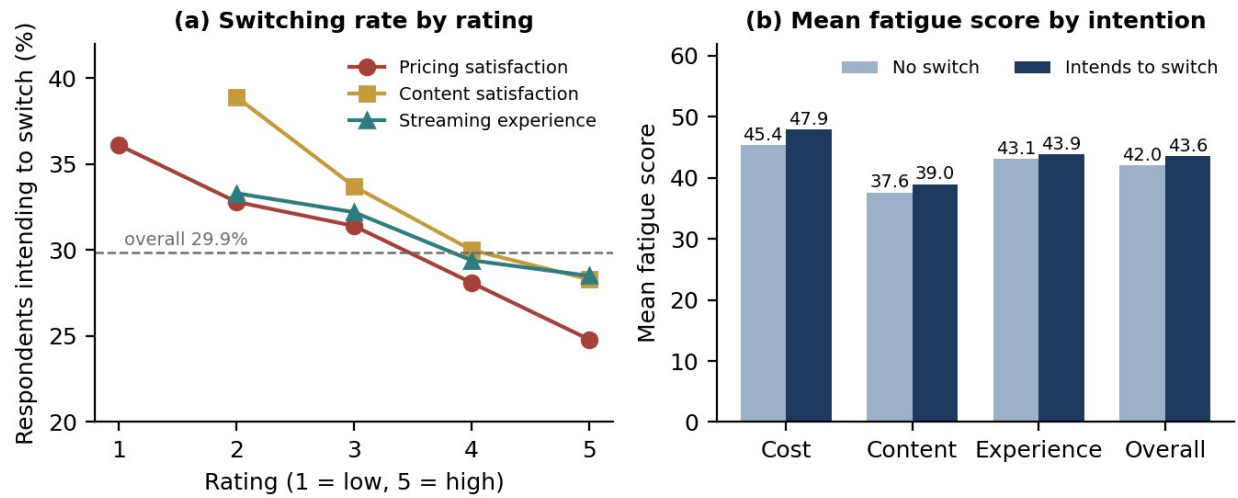


Figure 2. (a) Share intending to switch at each rating (approximate values read from the notebook cross-tabulation charts) and (b) mean fatigue score by switching intention (bars start at zero).

One caution from the correlation heatmap (Appendix A1): the six satisfaction ratings are almost uncorrelated with one another (between  $-0.02$  and  $0.02$ ). Ratings in real surveys normally move together, so this is most likely a side effect of how the synthetic rows were generated, and it is one more reason to read the findings as directional.

## 4. Analytics Method and Implementation

I used **logistic regression**, a supervised classification method, for three reasons. The target is binary (switch or not), which logistic regression models directly. Its coefficients show the direction and relative strength of each factor, so a retention team can see why a subscriber is flagged and not only that they are. And it is light enough to retrain regularly. Kirgiz et al. (2024) compared logistic regression with random forest on telecom churn and went with random forest for its predictive power. I faced the same trade-off and stayed with logistic regression because a flag has to be explainable to the people acting on it, and because a more flexible model would mostly learn the quirks of my largely synthetic rows.

### Implementation

1. Kept the 9,999 records with a target label and 25 predictors (16 numeric, 9 categorical).
2. Split the data 80:20 with stratification, giving 7,999 training and 2,000 test records, so both sets keep the same share of switchers.
3. Numeric variables: median imputation, then standardisation. Categorical variables: most-frequent imputation, then one-hot encoding. Both steps sit inside one scikit-learn pipeline fitted on the training data only, so nothing from the test set leaks into preprocessing.
4. Fitted logistic regression with balanced class weights, which weights each class inversely to its frequency so the 30% switcher group is not swamped by the majority (scikit-learn developers, n.d.).
5. Evaluated on the held-out test set with accuracy, precision, recall, F1 and the confusion matrix.

Table 4. Test-set performance (n = 2,000; 598 real switchers in the test set)

| Metric    | Result | What it means for retention                                                           |
|-----------|--------|---------------------------------------------------------------------------------------|
| Accuracy  | 62.05% | Share of all subscribers classified correctly                                         |
| Precision | 40.92% | Of the subscribers flagged as likely switchers, the share who really intend to switch |
| Recall    | 60.70% | Of the subscribers who intend to switch, the share the model catches                  |
| F1-score  | 48.89% | Balance between precision and recall                                                  |

|                     |                        |                     |
|---------------------|------------------------|---------------------|
| Actual<br>No Switch | 878                    | 524                 |
|                     | 235                    | 363                 |
|                     | Predicted<br>No Switch | Predicted<br>Switch |

Figure 3. Confusion matrix on the test set.

The model catches 363 of the 598 switchers in the test set and misses 235. It also raises 524 false alarms. That trade-off suits a screening task: losing a subscriber costs more than sending a low-cost nudge to someone who was not at risk. Overall the model flags 887 of 2,000 users (44%), and 41% of them are real switchers against 30% in the test set as a whole, a lift of about 1.4 over flagging at random. Accuracy (62.05%) is below the roughly 70% that always predicting “No Switch” would score, which is expected when class weights favour recall. The model is therefore a rough shortlisting tool, not a precise predictor. The notebook adds further checks (ROC-AUC, five-fold cross-validation, coefficients and a threshold sweep in Section 12.3) to test it more thoroughly.

## 5. Comparison with State-of-the-Art Methods

I compared my study with three recent published studies (Table 5). Their datasets, methods and metrics differ from mine, so the comparison looks at design, evidence and findings. I do not rank the studies by reported score.

Table 5. State-of-the-art comparison

| Published study / year | Dataset                                                                            | Method used                                                                                                                    | Evaluation metric                                            | Key result                                                                                                            | Comparison with my work                                                                                                                                                                                                                                                                            |
|------------------------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Anumolu et al. (2024)  | OTT customer churn data, processed at scale with Apache Spark                      | Machine-learning churn models on Spark; a hybrid approach compared with a non-hybrid baseline                                  | AUC                                                          | The hybrid approach reached a higher AUC than the non-hybrid baseline                                                 | They work with large-scale platform data; I use a small survey with stated intention. They focus on prediction accuracy; I add a fatigue diagnosis and keep one interpretable model. I report accuracy, precision, recall and F1 at one threshold, so my scores are not comparable with their AUC. |
| Kirgiz et al. (2024)   | Secondary data from a telecom operator on churner and non-churner mobile customers | Logistic regression compared with random forest; random forest chosen; importance by mean decrease in impurity and permutation | Predictive power of the two classifiers; variable importance | Gamified loyalty giveaways and branded apps were effective; OTT-service strategies mattered less for predicting churn | The closest method match: we both start from logistic regression, but they choose random forest for power and I keep logistic regression for explainability. They use observed churn labels; I use stated intention. Their setting is telecom, mine is OTT.                                        |

| Published study / year | Dataset                                         | Method used                                                                                                 | Evaluation metric                        | Key result                                                                                                                                                       | Comparison with my work                                                                                                                                                                                                                                                                                                              |
|------------------------|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jeong & Lee (2025)     | Survey of OTT multi-homing users in South Korea | Framework based on the stimulus-organism-response model and mental accounting theory, tested on survey data | Significance of the hypothesised effects | High search costs and fees drive discontinuation; content diversity and efficient recommendations reduce it; fatigue and mental coupling are significant drivers | Same kind of data (a questionnaire) and similar themes (cost, content, recommendations), which supports my cost, content and experience framing. They explain why users leave using genuine respondents; I predict who may leave, using mostly synthetic rows. My dominant fatigue type is a practical label, not a validated scale. |

The three studies sit on two sides of my work. Anumolu et al. and Kirgiz et al. predict churn from behavioural or account data, while Jeong and Lee explain discontinuation through cost, content and fatigue using survey data. My case study borrows from both: a classifier for who is at risk and a simple fatigue label for why. The strength of this design is the direct link from a prediction to a type-specific action. Its weaknesses are a simple model and largely synthetic data, so my scores carry less weight than theirs. This comparison does not show that my approach performs better or worse than any of the three.

## 6. Results, Business Insights and Recommendations

### Key results

- The analytical dataset has 10,000 records (13 real, 9,987 synthetic) and 32 fields, with no duplicates.
- 29.88% of respondents intend to switch within six months. The dominant fatigue type is cost for 44.25%, experience for 34.51% and content for 21.24%.
- Switching intention falls as pricing, content and streaming ratings rise, most steeply for pricing.
- The class-balanced logistic regression reaches 62.05% accuracy, 40.92% precision, 60.70% recall and 48.89% F1 (363 true positives, 524 false positives, 235 false negatives, 878 true negatives).

### Business insights

**Fatigue has more than one face.** Cost is the largest group, but 55.75% of respondents are mainly affected by experience or content problems that a price cut will not fix. A single retention lever would miss most of them.

**Pricing satisfaction is the clearest warning sign.** Respondents who rate pricing 1 or 2 intend to switch at roughly one in three, compared with about one in four among those who rate it 5. Content and streaming ratings move in the same direction with smaller gaps.

**Prior switching and competitor awareness are too common to separate people on their own.** About half of respondents have switched before (49.88%) and about half know of competing offers (48.92%). They are useful as tie-breakers on top of the model's risk score, not as filters.

**The model is a screening tool.** With about four in ten flagged users being real switchers, its output should decide who gets a look, not who gets an expensive offer.

## Recommendations by fatigue type

Table 6. Fatigue-to-action mapping

| Fatigue type    | Key signals                                                            | Recommended action                                                                                                | Aim                             |
|-----------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------|
| Cost            | Low pricing satisfaction; high spend; several concurrent subscriptions | Value plans such as student or family bundles, a cheaper ad-supported tier, usage-linked offers                   | Raise perceived value for money |
| Content         | Low content satisfaction or variety rating                             | Better genre-level personalisation, curated collections, catalogue additions guided by the open-text answers      | Improve catalogue fit           |
| Experience      | Low streaming, recommendation or advertisement-tolerance ratings       | Fix buffering and playback stability, app performance and recommendation quality; review ad frequency on ad tiers | Remove everyday friction        |
| Mixed / unclear | High predicted risk but no standout fatigue type                       | Manual review or a short satisfaction survey before any offer is made                                             | Avoid one-size-fits-all offers  |

In practice I would score subscribers monthly, attach each one's dominant fatigue type and match it to Table 6. Because most flagged users are not true switchers, first-line actions should be cheap and low-friction: a satisfaction prompt, better recommendations or a small loyalty offer. Larger discounts should be kept for users who combine a high predicted risk with clear cost fatigue. Results should be measured against a holdout group rather than by before-and-after comparison, precision and recall should be monitored as behaviour drifts, and the model should be retrained on actual cancellation outcomes once enough have accumulated.

## 7. Conclusion and References

This case study followed a complete analytics workflow on OTT subscription fatigue. I framed a business problem, collected questionnaire data, prepared and explored it, diagnosed fatigue type, built a logistic regression model for six-month switching intention, compared the approach with three recent studies, and turned the findings into segment-specific actions.

On the analytical dataset, the class-balanced model caught 60.7% of would-be switchers at 40.9% precision, which is useful for shortlisting but not for automatic high-cost decisions. Switching intention was higher among users with lower pricing, content and streaming ratings, and cost was the most common dominant fatigue type (44.25%), although more than half of respondents were mainly bothered by content or experience. The main caveat is the data: only 13 of the 10,000 records are real responses. The next step is to collect more genuine responses and repeat the analysis. If the patterns hold, the mapping in Table 6 is ready to be piloted.

### Limitations

- **Data.** Only 13 records are real. The synthetic rows follow patterns from that tiny sample, the satisfaction ratings are almost uncorrelated with each other, and the respondent profile is young and student-heavy. The findings need re-testing on genuine responses.
- **Stated intention.** The target is what respondents say they will do, not observed cancellations, and the two can differ.
- **Association, not causation.** The analysis does not show that changing price, content or experience would change switching behaviour.
- **Overlapping predictors.** The fatigue scores are built from the same ratings that also enter the model, so individual coefficients should be read with care. The three scores are also on different ranges, so the order of the dominant fatigue types is approximate. The notebook tests a model without the fatigue scores.

- **One algorithm and one threshold.** I evaluated a single model at the default 0.5 cut-off. The fatigue mix among people who intend to switch is also still to be reported; the notebook contains the cell for it.

## References

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[https://scikit-learn.org/stable/modules/generated/sklearn.linear\\_model.LogisticRegression.html](https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LogisticRegression.html)

## Appendix A1. Correlation heatmap

Output of the notebook (Section 9.9) for the numeric, rating and fatigue variables. The strong negative values between a fatigue score and its ratings show how the scores are constructed. The near-zero values among the six satisfaction ratings are discussed in Section 3.

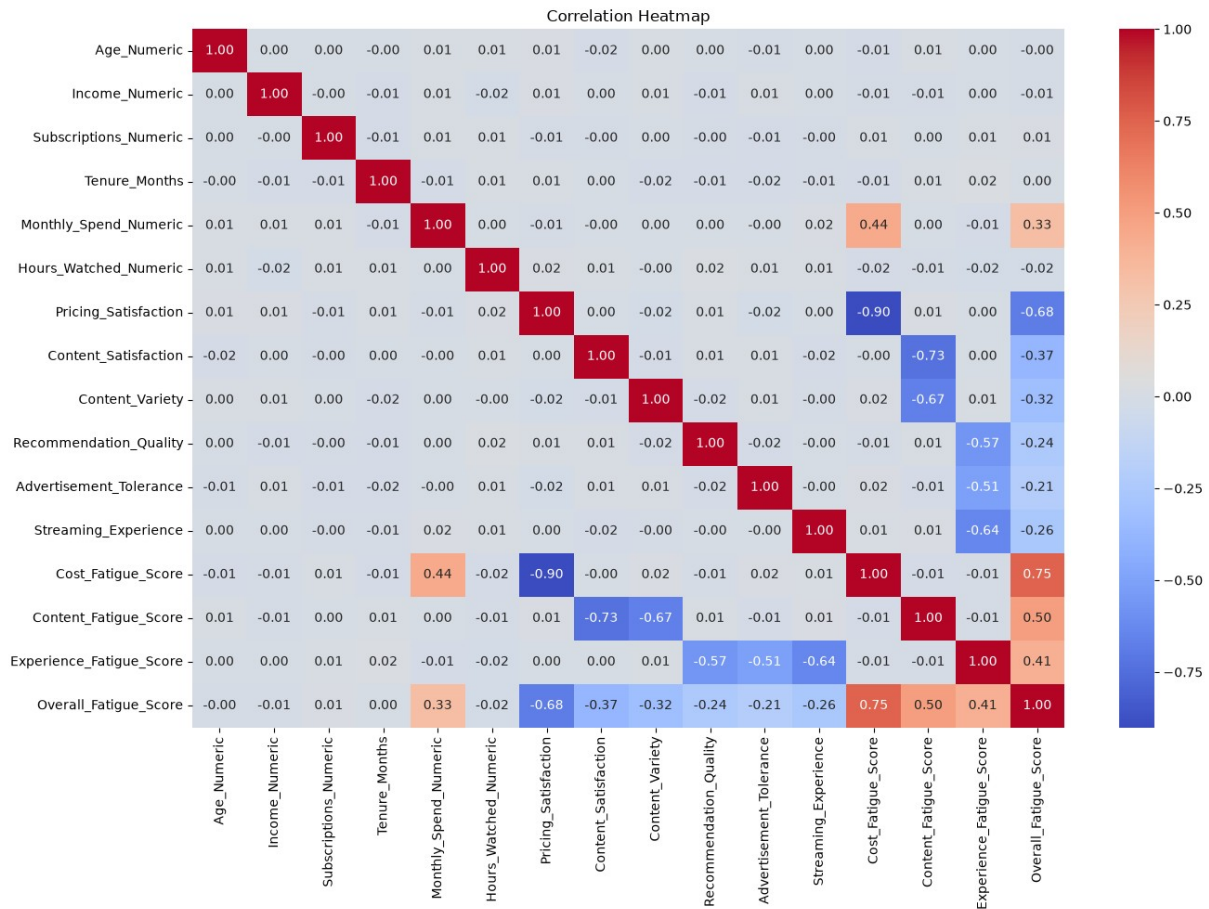


Figure A1. Correlation heatmap (10,000 records).